

Indian / Bollywood Version of Connect the Stars

End-to-End Implementation Plan

Refined Prompt

Create an end-to-end implementation plan for an Indian/Bollywood version of Connect the Stars. The product should work like a movie trivia web game where players connect two actors by entering movies and co-actors, but the theme, dataset, design, content, and growth strategy should be optimized for Bollywood and Indian cinema.

1. Product Concept

Build a daily and endless Bollywood trivia web game where players connect two Indian film stars through shared movies. The player starts with one actor and must reach the target actor by alternating between movies and actors.

Example Path	Meaning
Shah Rukh Khan -> Dear Zindagi -> Alia Bhatt	Direct connection through a shared movie
Aamir Khan -> Dil Chahta Hai -> Saif Ali Khan -> Kal Ho Naa Ho -> Shah Rukh Khan -> Dear Zindagi -> Alia Bhatt	Multi-step connection through co-actors and films

2. Suggested Product Names

- Star Jod
- Bollywood Link
- Filmy Connection
- Connect Bollywood Stars
- CineLink India
- Filmy Chain

3. MVP Scope

- Bollywood-only actor and movie graph for version 1.
- Random actor pair generator.
- Manual actor pair selection.
- Autocomplete search for movies and actors.
- Actor -> movie -> actor validation.
- Hints, undo, reset, and share result.
- Daily Bollywood challenge.
- Leaderboard based on shortest path and completion time.
- Difficulty modes: Easy, Medium, Hard, Legend.

4. Data Source Strategy

Use TMDB as the main data source for actor, movie, cast, poster, and profile-image metadata. Use IMDb non-commercial datasets only for internal experimentation unless proper licensing is secured for commercial usage.

TMDB API / optional licensed datasets



Data ingestion worker



Normalize movies, actors, cast



Create actor-movie graph



Store in PostgreSQL + Redis cache



Game API uses local database

5. Database Schema

Core Tables

```
CREATE TABLE actors (  
  id SERIAL PRIMARY KEY,  
  tmdb_id INT UNIQUE,  
  imdb_id TEXT,  
  name TEXT NOT NULL,  
  normalized_name TEXT NOT NULL,  
  gender TEXT,  
  profile_image_url TEXT,  
  popularity_score NUMERIC DEFAULT 0,  
  is_bollywood BOOLEAN DEFAULT FALSE,  
  is_active BOOLEAN DEFAULT TRUE,  
  created_at TIMESTAMP DEFAULT NOW()  
);  
  
CREATE TABLE movies (  
  id SERIAL PRIMARY KEY,  
  tmdb_id INT UNIQUE,  
  imdb_id TEXT,  
  title TEXT NOT NULL,  
  normalized_title TEXT NOT NULL,  
  release_year INT,  
  original_language TEXT,  
  poster_url TEXT,  
  popularity_score NUMERIC DEFAULT 0,  
  is_hindi BOOLEAN DEFAULT FALSE,  
  is_bollywood BOOLEAN DEFAULT FALSE,  
  is_documentary BOOLEAN DEFAULT FALSE,  
  created_at TIMESTAMP DEFAULT NOW()  
);  
  
CREATE TABLE movie_cast (  
  movie_id INT REFERENCES movies(id),  
  actor_id INT REFERENCES actors(id),  
  character_name TEXT,  
  billing_order INT,
```

```
PRIMARY KEY (movie_id, actor_id)
);
```

Aliases and Games

```
CREATE TABLE aliases (
  id SERIAL PRIMARY KEY,
  entity_type TEXT CHECK (entity_type IN ('actor', 'movie')),
  entity_id INT NOT NULL,
  alias TEXT NOT NULL,
  normalized_alias TEXT NOT NULL
);
```

```
CREATE TABLE games (
  id UUID PRIMARY KEY,
  start_actor_id INT REFERENCES actors(id),
  target_actor_id INT REFERENCES actors(id),
  difficulty TEXT,
  status TEXT DEFAULT 'active',
  created_at TIMESTAMP DEFAULT NOW(),
  completed_at TIMESTAMP
);
```

```
CREATE TABLE game_moves (
  id SERIAL PRIMARY KEY,
  game_id UUID REFERENCES games(id),
  move_number INT,
  entity_type TEXT CHECK (entity_type IN ('actor', 'movie')),
  entity_id INT NOT NULL,
  created_at TIMESTAMP DEFAULT NOW()
);
```

6. Game and Graph Logic

The game graph is bipartite: actors connect to movies, and movies connect to actors. Players must alternate between actor and movie nodes.

```
Actor <-> Movie <-> Actor <-> Movie <-> Actor
```

```
function isValidMove(currentNode, nextNode) {
  if (currentNode.type === "actor" && nextNode.type === "movie") {
    return movieCastExists(nextNode.id, currentNode.id);
  }

  if (currentNode.type === "movie" && nextNode.type === "actor") {
    return movieCastExists(currentNode.id, nextNode.id);
  }

  return false;
}
```

```
}
```

Winning Condition

```
if (nextNode.type === "actor" && nextNode.id === targetActorId) {  
  markGameCompleted();  
}
```

7. Shortest Path and Hint Engine

Use BFS for MVP. Later, move adjacency lists to Redis for faster hint generation and pair validation.

```
function findShortestPath(startActorId, targetActorId) {  
  const queue = [{ type: "actor", id: startActorId }];  
  const visited = new Set();  
  
  while (queue.length > 0) {  
    const path = queue.shift();  
    const current = path[path.length - 1];  
  
    if (current.type === "actor" && current.id === targetActorId) {  
      return path;  
    }  
  
    const neighbors = getNeighbors(current);  
    for (const neighbor of neighbors) {  
      const key = `${neighbor.type}:${neighbor.id}`;  
      if (!visited.has(key)) {  
        visited.add(key);  
        queue.push([...path, neighbor]);  
      }  
    }  
  }  
  return null;  
}
```

Hint Type	Example
Soft clue	This actor worked in a 2012 romantic drama.
First-letter clue	The movie starts with J.
Poster silhouette	Show blurred movie poster.
Best next step	Reveal one optimal actor/movie move.
Decade clue	This connection is from a 1990s film.

8. Search and Autocomplete

Bollywood search must handle nicknames, acronyms, alternate spellings, punctuation, and common short forms.

```
function normalizeText(value: string) {  
  return value  
    .toLowerCase()
```

```

.normalize("NFD")
.replace(/[\u0300-\u036f]/g, "")
.replace(/[^a-z0-9 ]/g, "")
.replace(/\s+/g, " ")
.trim();
}

```

Alias	Canonical Entity
SRK	Shah Rukh Khan
Shahrukh Khan	Shah Rukh Khan
Big B	Amitabh Bachchan
Bhaijaan	Salman Khan
K3G	Kabhi Khushi Kabhie Gham
DDLJ	Dilwale Dulhania Le Jayenge

Search ranking priority: exact alias match > exact title/name > starts-with match > fuzzy match > popularity score.

9. Frontend Implementation

Area	Recommendation
Framework	Next.js + TypeScript
Styling	Tailwind CSS
Animation	Framer Motion
Graph UI	React Flow or Cytoscape.js
State	Zustand
API cache	TanStack Query

Main Pages

```

/           Landing page
/play       Game screen
/play/[gameId]  Active game
/daily      Daily challenge
/leaderboard  Leaderboard
/about       Rules and credits
/admin       Admin dashboard

```

Visual Direction

- Dark cinematic background
- Gold and red-carpet accents
- Film reel and spotlight effects
- Subtle star particles
- Poster-card style for movies
- Premium Filmfare + IMDb + Wordle vibe

10. Backend Implementation

Component	Recommendation
Backend	Node.js with Fastify or NestJS
ORM	Prisma
Database	PostgreSQL
Cache	Redis

Jobs	BullMQ
Data API	TMDB integration
Monitoring	Sentry + CloudWatch/logging

API Endpoints

```

POST /api/games
GET /api/games/:gameId
POST /api/games/:gameId/move
POST /api/games/:gameId/undo
POST /api/games/:gameId/hint
GET /api/search?q=srk
GET /api/actors/:id
GET /api/movies/:id
GET /api/leaderboard
POST /api/leaderboard
GET /api/daily
POST /api/daily/submit

```

11. Data Ingestion Pipeline

1. Seed 100-300 popular Bollywood actors manually.
2. Fetch TMDB person records for each actor.
3. Fetch each actor's movie credits.
4. Filter for Indian/Hindi/Bollywood films.
5. Fetch full movie credits for approved movies.
6. Insert actors, movies, and movie_cast relationships.
7. Generate aliases and normalized search fields.
8. Build Redis adjacency lists for fast graph traversal.
9. Queue uncertain entities for admin review.

```

actor:{actorId}:movies -> [movieId1, movieId2, movieId3]
movie:{movieId}:actors -> [actorId1, actorId2, actorId3]

```

12. Difficulty Algorithm

Difficulty	Actor Pool	Shortest Path Target	Example
Easy	Top 100 actors	1-3 edges	SRK -> Dear Zindagi -> Alia
Medium	Top 300 actors	3-5 edges	Mainstream co-actor chain
Hard	Top 1000 actors	5-7 edges	Character actors and older films
Legend	Large historical pool	7+ edges	Obscure classics and supporting actors

```

async function generatePair(difficulty) {
  const actors = await getActorsByDifficultyPool(difficulty);

  while (true) {
    const start = randomItem(actors);
    const target = randomItem(actors);
    if (start.id === target.id) continue;

```

```

const path = await findShortestPath(start.id, target.id);
if (isPathValidForDifficulty(path, difficulty)) {
  return { start, target, shortestPathLength: path.length };
}
}
}

```

13. Game Modes

Mode	Description
Classic	Connect two random Bollywood actors.
Daily Challenge	Same puzzle for everyone each day.
Speedrun	Fastest completion wins.
Shortest Path	Fewest moves wins.
Party Mode	One shared screen for groups.
Regional Mode	Hindi, Marathi, Tamil, Telugu, Malayalam, Kannada, Bengali.
Movie-to-Movie	Connect one film to another through actors.
Theme Mode	90s, villains, romance, YRF, Dharma, comedy, classics.

14. Admin Panel

- Approve/reject actors and movies
- Merge duplicate actors
- Add aliases
- Mark movie as Bollywood/Indian/regional
- Remove documentaries or irrelevant movies
- Fix wrong cast data
- Create daily challenge
- View failed search terms
- View most requested actors

```

CREATE TABLE moderation_queue (
  id SERIAL PRIMARY KEY,
  entity_type TEXT,
  entity_id INT,
  issue_type TEXT,
  status TEXT DEFAULT 'pending',
  notes TEXT,
  created_at TIMESTAMP DEFAULT NOW()
);

```

15. Deployment Architecture

MVP	Production
Vercel frontend	Cloudflare CDN + Vercel/Next.js
Render/Railway/Fly.io API	AWS ECS/Fargate or Lambda API
Supabase/Neon Postgres	AWS RDS PostgreSQL
Upstash Redis	ElastiCache or Upstash Redis
Cloudinary/S3 assets	S3 + CloudFront
Basic logs	Sentry + CloudWatch + analytics

16. Suggested Folder Structure

```
bollywood-connect/
├── apps/
│   ├── web/
│   │   ├── app/
│   │   ├── components/
│   │   ├── hooks/
│   │   ├── lib/
│   │   └── styles/
│   └── api/
│       ├── src/
│       │   ├── modules/
│       │   │   ├── game/
│       │   │   ├── search/
│       │   │   ├── actors/
│       │   │   ├── movies/
│       │   │   ├── leaderboard/
│       │   │   └── admin/
│       │   ├── jobs/
│       │   ├── services/
│       │   └── main.ts
│       └── packages/
│           ├── db/
│           ├── shared/
│           └── ui/
├── scripts/
│   ├── ingest-tmdb.ts
│   ├── build-graph.ts
│   ├── generate-daily.ts
│   └── normalize-aliases.ts
└── README.md
```

17. Six-Week Build Roadmap

Week	Focus	Deliverables
1	Product and setup	Name, wireframes, repo, database, TMDB key, seed actors
2	Data pipeline	Actor fetch, movie credits, cast graph, normalization, aliases
3	Game engine	Create game, validate moves, BFS, win condition, undo/reset
4	UI	Landing page, game screen, graph, autocomplete, hints, mobile layout
5	Leaderboard and daily	Daily puzzle, scoring, share cards, analytics
6	Polish and launch	Admin panel, bug fixes, SEO, deployment, soft launch

18. Monetization and Growth

Idea	Implementation
Ads	Show ads after game completion or between challenges.
Premium	No ads, unlimited hints, custom actor pairs, private party rooms.
Sponsored Challenges	YRF Week, SRK Birthday Challenge, 90s Challenge.
Creator Mode	Influencers and film pages create custom challenges.
College/Party Mode	Paid private rooms for events and quizzes.

Share Card Example

I connected Shah Rukh Khan -> Alia Bhatt in 2 moves!

★ Shah Rukh Khan
🎬 Dear Zindagi
★ Alia Bhatt

Can you beat me?

19. Legal and Data Notes

- Do not scrape IMDb pages directly. Use official APIs or licensed datasets only.
- Check TMDb attribution and image usage requirements before launch.
- Do not copy the reference website's branding, UI, or assets.
- Use your own Bollywood-inspired product identity.
- Maintain an admin moderation process because movie/cast metadata can be messy.

20. Final Recommended Architecture

Frontend: Next.js + TypeScript + Tailwind + React Flow
Backend: Node.js/Fastify or NestJS
Database: PostgreSQL + Prisma
Cache: Redis
Search: PostgreSQL pg_trgm first, Meilisearch later
Data: TMDb API -> ingestion worker -> local database
Deployment: Vercel + Supabase/Neon + Upstash for MVP
Scale: AWS ECS/RDS/ElastiCache/S3/CloudFront later

21. One-Line Product Definition

A daily and endless Bollywood trivia web game where players connect two Indian film stars through shared movies, with hints, leaderboards, shareable results, and regional cinema expansion.